

① A company uses an autonomous robot to deliver medicines within a hospital. The robot navigates through corridors, avoids obstacles, identifies patient rooms, and delivers medicines to the correct location.

(i) Construct the PEAS description for this intelligent agent.

PEAS is a foundational framework used to specify, design and evaluate the task environment of an intelligent agent. Here is the PEAS description for the Autonomous Hospital Medicine Delivery Robot:

PERFORMANCE MONITOR (P): Delivers medicines to the correct patient room, minimize delivery time, avoid collisions, ensure safety, conserve battery power, complete tasks successfully.

ENVIRONMENT (E): Hospital corridors, patient rooms, nurses, doctors, patients, elevators, walls, doors, and moving obstacles.

ACTUATORS (A): Wheels/motors for movement, robotic arm or medicine dispenser, door/elevator control interface, speaker or display for communication.

SENSORS (S): Cameras, LiDAR, ultrasonic sensors, GPS/indoor localization sensors, RFID/barcode scanners, proximity sensors, DMR Sensors.

(ii) Identify the environment type of this agent.

The agent operates in a Partially Observable, Dynamic and Stochastic environment.

① Partially Observable

- i) The robot cannot always observe the entire hospital environment.
- ii) Obstacles may suddenly appear around corners or behind doors.

② Dynamic

- i) The environment changes while the agent is operating.
- ii) People, wheelchairs, and equipment move continuously.

③ Stochastic

- i) Outcomes are uncertain.
- ii) Sensors may have errors & human movement is unpredictable.

② Why do Goal-Based Agents often require search & planning techniques? Explain with an example.

A goal-based agent chooses actions that help achieve a specific goal. To reach the goal, it must determine the sequence of actions that is best suitable from the current state to the desired state.

Significance of "Search"

- i) There may be many possible actions.
- ii) The agent searches through possible states, finding a path to the goal.

Significance of "Planning"

- i) The agent must organize actions in the correct order.
- ii) Planning helps minimize cost, time, or distance.

Example

Consider a robot whose goal is to deliver medicine from the pharmacy to room 408.

Goal: Reach room 408.

Search: Find possible routes

Planning: Select the shortest & safest route avoiding crowded areas & obstacles.

Thus, goal-based agents use "Search" to find possible solutions and "Planning" to choose & execute the best one.

③ A company develops a virtual assistant that can understand spoken commands such as "Set an alarm for 7 AM" and answer customer questions in English. Which branch(es) of AI are involved in developing this system?

① Natural Language Processing (NLP)

- i) Allows the system to interpret human language.
- ii) Extract the user's intent from spoken commands.

② Speech Recognition (SR)

- i) Converts spoken words into text.
- ii) Allows users to interact using voice commands.

③ Machine Learning (ML)

- i) Improves understanding and response quality through training on large datasets.
- ii) Helps recognize different accents & question patterns.